© State Examinations Commission

Question 9 **(Suggested maximum time: 5 minutes)**

(i) Fill in the first difference and the second difference for the following patterns.
Some of Pattern 1 has been completed for you.

Pattern 1: 2 4 9 17 28

First difference: (*Change*) 2 **5** 8 11

Second difference: (*Change of change*) 3 3 **3**

Pattern 2: –1 2 5 8 11

First difference: (*Change*) 3 3 3 3

Second difference: (*Change of change*) 0 0 0

(ii) State whether each pattern is **linear** or **quadratic**. Give a reason for each answer.

Pattern 1: Quadratic

Reason: First different changes, 2nd difference is constant

Pattern 2: Linear

Reason: First difference is constant.

Excerpt provided by
www.themathstutor.ie